



Faculty of Engineering and Technology

Computer Science Department

USABLE SECURITY AND PRIVACY

COMP 438

**Critical Analysis of “On Privacy and Security
Challenges in Smart Connected Homes”**

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Limitations of the Proposed Research Paper

As reading and analyzing this research paper, limitations have been discovered that can be clearly pointed out, these limitations have taken into consideration fairness but need to be discussed clearly.

One limitation from this paper that can be shown is that this paper mainly discusses how smart homes are attached to the idea of privacy and security problems and how they affect the people in the world but is limited in showing real world examples or experiences with users who have smart homes, it doesn't provide the user feedback on how these smart homes can affect their privacy and security, it gives us mainly the discussion that it does.

Another limitation upon reading and analyzing this research paper is that it focused more on the technical challenges more than on how users can be affected and or what their experience can be with smart connected homes, this paper does explain the device, communication and service level in a good and clear way but doesn't show how users can be affected or how normal users understand the privacy risks or how to manage these risks on having smart connected homes, people who may have smart homes can be of different ages or different technical skills, not taking these two factors into consideration shows the paper is limited in this area.

When coming to the problems and security risks section, it gives the idea on the challenges and generally explains them with identifying the risk and assessing it as well, but doesn't dive deep into how these can affect the users with smart homes and doesn't really discuss how users can prioritize these risks such as explaining which risk can be the most dangerous or important to take into consideration, without this the paper can limit other researchers on not knowing where to focus on creating their own research papers based on security risks for smart homes.

Proposed Improvements

One of the things that has space for improvement would be to do real world testing and help reduce the lack of not having the user experience and feedback, this will help other researchers and people who are doing their research on smart homes to have an idea on the user experience and the usability of these smart homes. Researchers can test the security and privacy solutions proposed using real devices and real people. This will help determine if these solutions are reasonable for our daily life and not just theories.

Future studies can include surveys, interviews with users or testing to understand the user perspective of secure and private and how these devices can affect user experience based on different factors such as user interface for these devices.

Looking into the limitations, the study could improve on making it clear to the users the level of risk and challenges that can arise, researcher or author could help determine which problems have a higher risk on security and privacy.